

L14 Team 02

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mNICU Project Critical Design Report

Executive Summary

This report presents the design, analysis, and verification process of retrofitting an existing air handling unit (AHU) for a mobile neonatal intensive care unit (mNICU). This project is based on engineering requirements given by the client specifying the controlled volumetric flow rate and temperature of the air to be delivered, as well as the system's reaction to any disturbances. As described in the results section, each of these requirements were met and confirmed during the final verification testing.

The first step for the design process involved experimental data acquisition and calibration of the existing AHU. Differential pressure transducers were first calibrated against reference manometers. Next, thermocouple selection was done by measuring the dynamic response and accuracy of various types in order to replace the costly temperature sensor used previously. Then, an orifice plate flow meter was designed to ASME MFC-3M-2004 standards and simulated using Finite Element Analysis to resist the expected maximum pressure. After this, fan performance was characterized using dimensional analysis and similitude, and heater power requirements were calculated using an energy balance and predicted resistance values from each component of the AHU. Finally, a proportional-integral (PI) controller was tuned experimentally to meet the settling time requirements in case of a disturbance in air flow.

Key results found during verification testing show that the measured flow rate of $0.0343 \pm 1.47\%$ m³/s, meeting the client's requirement of $0.034 \pm 5\%$ m³/s, the air temperature is controlled at the required 23°C, and the settling time in case of disturbance is ≤ 76 seconds, measuring between 45-66 seconds depending on the type of disturbance. The system demonstrated stable and controlled behavior under disturbance with no manual adjustment necessary during verification testing.

Recommended next steps include the planning and manufacturing of aluminum orifice plates to ensure longevity and efficiency of the system. While the 3D-printed PLA orifice plates were shown to be effective and able to resist the expected pressures, they are more vulnerable to the potential heat of the air through the AHU and would not last as well as an aluminum or steel design. Another option for further improvement is more realistic modeling of the system in order to dial in the PI control further. While not necessary under the client's requirement, a faster response time is optimal.

Introduction

Throughout this project, our team worked to retrofit our company's small air handling unit (AHU) to function in a mobile neonatal intensive care unit (mNICU). The main requirements from our client is to provide a steady volume flow rate of air at a controlled temperature, while responding to any sudden disturbances. The system must supply air at a volumetric flow rate of $0.034 \pm 5\%$ m³/s, corresponding to an air change rate of $9 \pm 5\%$ ACH. The air temperature supplied to the mNICU must be maintained at $23 \pm 1^\circ\text{C}$, with the control system achieving a settling time of no more than 76 seconds following a disturbance.

To meet these requirements, each pressure transducer must first be calibrated against a standard manometer, the heater and fan performance as well as ducting losses must be characterized for thermodynamic energy balance, an ASME MFC-3M-2004 compliant orifice plate flow meter

must be designed and manufactured via 3D printing, and finally a proportional-integral control temperature controller must be calibrated for a quick and stable reaction to disturbances. The project is divided into two main phases, first modeling the existing AHU, then using this model and energy balance calculations to meet the design requirements. Additionally, an orifice plate was designed for proper measurement of the volumetric flow rate at the outlet.

Analysis or Methodologies

The final AHU design was developed through an iterative engineering process over the course of 11 weeks, integrating experimental measurements, analytical modeling, and verification testing to ensure all mNICU engineering requirements were met. The process began with establishing accurate measurement techniques through sensor calibration, followed by component level design of the flow meter, thermal system, and fan performance, and concluded with system level validation through control testing and verification.

Accurate data collection was ensured through calibration of differential pressure transducers using manometers as reference standards. Known pressure differences were applied using a precision air pump, and the corresponding voltage outputs were recorded to develop linear calibration relationships between voltage and pressure. This allowed determination of sensitivity and bias for each transducer, which were then used to convert all measured voltages into pressure values throughout the project. Calibration was performed over both low and high pressure ranges to ensure appropriate sensitivity for different operating conditions. This step was critical because all subsequent flow rate and energy calculations relied on these calibrated measurements, and minimizing calibration error ensured reliable results.

Thermocouple selection was based on both accuracy and dynamic response requirements. Temperature data from multiple thermocouples were analyzed by plotting temperature versus time and evaluating the natural log of relative dynamic error to determine time constants. Smaller thermocouples demonstrated faster response due to reduced thermal inertia, and only the 1/16 inch thermocouple met the required settling time constraint for the system. Accuracy requirements further guided selection, as Type-T thermocouples provided tighter uncertainty bounds than Type-K thermocouples. Based on these considerations, a 1/16 inch grounded Type-T thermocouple was selected because it satisfied both the $\pm 1^\circ\text{C}$ accuracy requirement and the dynamic response requirement while minimizing disturbance to the airflow.

The flow meter design was based on an orifice plate developed using ASME MFC-3M-2004 standards. The beta ratio was selected iteratively to balance pressure drop, measurement sensitivity, and uncertainty, ensuring operation within acceptable Reynolds number limits and transducer measurement ranges. The final beta value provided sufficient pressure differential while maintaining low flow measurement uncertainty. Geometric parameters such as plate diameter, thickness, and bevel angle were selected according to standard guidelines to minimize flow separation and ensure accurate measurements. Finite Element Analysis (FEA) was conducted to verify structural integrity of the plate under operating pressure conditions, evaluating both maximum displacement and Von Mises stress. Results confirmed that the design remained well within allowable limits, with a high factor of safety, ensuring that deformation would not impact measurement accuracy. The final design was then fabricated using 3D printing, allowing precise implementation of the geometry and direct integration into the AHU for testing.

Fan performance was analyzed using dimensional analysis and similitude principles. Experimental data were used to generate dimensional fan performance maps by plotting pressure rise versus volumetric flow rate, allowing determination of system resistance. Dimensionless analysis was then performed using flow and pressure coefficients, enabling prediction of fan performance at different operating conditions. By combining the system resistance curve with fan performance data, the required fan speed to achieve the design flow rate was predicted. This approach allowed accurate determination of operating conditions without the need to test every possible fan speed.

Heater performance and power requirements were evaluated using an energy balance based on the First Law of Thermodynamics. Mass flow rate was determined from flow measurements, and air density was calculated using the ideal gas law. The rate of heat input from the heater, fan work, and the change in air enthalpy were used to calculate heat loss within the system. Additional losses from ducting and AHU components were included to determine the total required heater power. This analysis confirmed that the heater provided sufficient energy to meet the required supply air temperature under expected operating conditions, and uncertainty analysis was performed to ensure the reliability of these calculations.

The control system was developed and tested using a proportional integral (PI) controller. Controller gains were tuned experimentally by adjusting proportional gain and integral time constant to achieve a balance between response speed, stability, and steady-state error. Initial high gains resulted in oscillatory, marginally stable behavior, while reducing gains improved stability but slowed response. Final gains were selected through iterative testing to meet settling time and accuracy requirements. Verification testing was conducted by introducing perturbations through opening and closing the bypass gate, simulating real operating disturbances. The system response was evaluated based on its ability to maintain stability and quickly return to the desired temperature setpoint. Final testing confirmed that the controller met performance requirements and maintained stable operation under both steady-state and transient conditions.

Results

The final design of the mNICU air supply system meets all specified engineering requirements and demonstrates stable results under verification testing. The recommended design includes an orifice plate flow meter with a diameter ratio of 0.5915, a discharge coefficient of 0.6173, using D-D/2 pressure taps, which conform to the ASME Standard. The control system was tuned to a proportional gain of 3.75, an integral gain of 18.75, and an integral time constant of 0.2 minutes. The AHU operates with an overall system resistance of 426194 kg/m^7 , a fan speed of 11750 rpm, a pressure difference across the fan of 486.90 Pa, a minimum ambient temperature of 16.83°C , and a reference temperature of 40.7°C .

Table 1 summarizes the design requirements and the results of verification testing. The required air change rate of $9 \pm 5\%$ ACH was achieved with a measured value of 9.051 ACH. The required volumetric flow rate of $0.034 \pm 5\%$ m^3/s was achieved with a measured value of $0.0343 \pm 1.47\%$ m^3/s . The real-time, in-line air flow measurement system met the accuracy requirement of $\leq \pm 4\%$ with an error of 3.43%. Temperature control was also successful, with the required air

temperature of $23 \pm 1^\circ\text{C}$ met with a measured air temperature of 23°C . The system met the settling time requirement of ≤ 76 seconds, between both control system verification tests, observed settling time ranged from 45 to 66 seconds.

Table 1. Design requirements for mNICU supply air system and results of verification testing

Customer Requirement	Engineering Requirement	Detailed Engineering	Verification Results
Required air flow (ACH)	$9 \pm 5\%$	Fan speed setting	9.051
Required air flow (m^3/s)	$0.034 \pm 5\%$	Fan speed setting	$0.0343 \pm 1.47\%$
Maximum flow (m^3/s)	0.09	Fan speed setting	0.0348
Real-time, in-line airflow measurement	$\leq \pm 4\%$ accuracy	Orifice plate geometry transducer specs	3.43%
Orifice plate outer diameter (mm)	102.36	ASME Standard	-
Orifice plate pressure taps	D-D/2	Orifice plate geometry	-
Orifice plate maximum thickness (mm)	≤ 3.6	ASME Standard	-
Orifice plate bevel angle ($^\circ$)	50	ASME Standard	-
Orifice plate manufacturing	3D print	Fabrication method	-
Orifice plate material	Polylactic acid	Material	-
Air temperature ($^\circ\text{C}$)	23 ± 1	Fan speed & heater power	23
Settling time (seconds)	≤ 76	P and I gains	45-66

No adjustments to the fan speed or control gains were necessary during verification testing because, as seen in Figure 1, the AHU operation remained stable when being perturbed, proving that the chosen control system design is fitting.

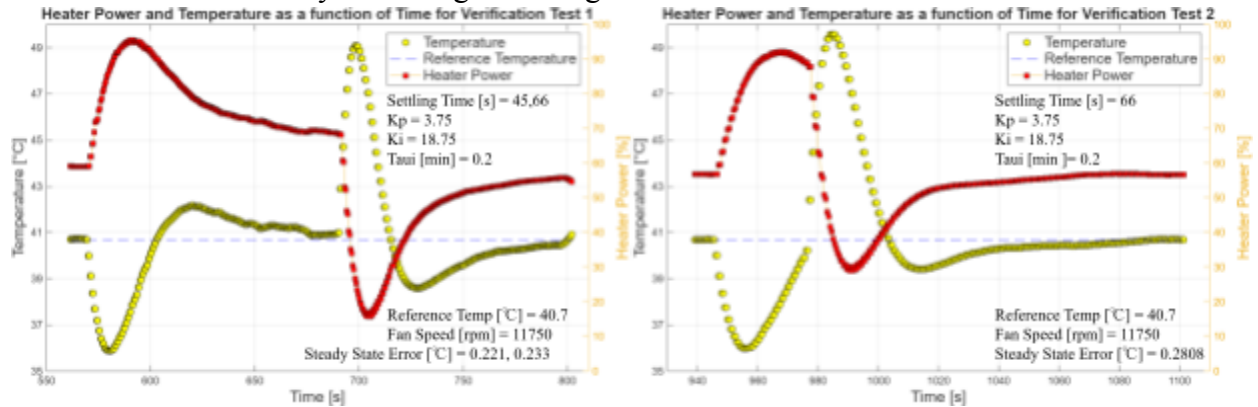


Figure 1. Steady-state operation for both tests of control system verification

The measured volume flowrate is slightly greater than the required volume flowrate. Both the MAF sensor and the orifice plate located on the main leg of the AHU measured consistent flow rates which aligned closely with the required value of $0.034 \text{ m}^3/\text{s}$. The orifice plate on the exit leg (from design phase 1) showed deviation from the required flow, which could be a result of installation effects, rounding errors, or sensitivity to measurement error. These discrepancies fall within the uncertainty bounds and do not cause a significant impact on the performance of the system. Measurement uncertainty in the installed orifice plates is mainly due to geometric tolerancing in the 3D-printed plate, but the measured flow accuracy falls within the $\pm 4\%$ requirement, proving that the design is effective.

Figure 1 shows the steady state operation for both control system verification tests, where the system is perturbed. Each plot shows the temperature and heater power, demonstrating stable system behavior. The system returns to steady state within the specified settling time with minimal steady state error.

Recommendations and Next Steps

In the end the mNICU AHU met all of the set engineering requirements quite well, with a settling time between 45 to 60 seconds and a volumetric flow rate of 0.0343 meters cubed per second. Because of the success of the orifice plate, the next step would be to create a plan for, and start machining orifice plates out of aluminum or steel to ensure longevity and efficiency of the system. Additionally small tweaks could be made to the PID controls to improve the response time and accuracy of the AHU even further.

Cost efficiency played a substantial role in our final design for the mNICU AHU, through balancing both costs with necessary accuracy. RTD thermo couples were turned down in favor of the Type T couples due to their significantly smaller price while still achieving the required $\pm 1^{\circ}\text{C}$ accuracy and quick response time. ISA nozzles were also turned down despite their superior flow control in favor of simplistic orifice plates, allowing prototypes to be easily printed and the opportunity for cheap manufacturing in the long run.

Keeping a consistent and accurate air temperature is crucial as infants who need the mNICU AHU can not yet regulate their own body temperature. If temperatures are too high the baby risks suffering from dehydration and heat stroke. On the opposite side, if the air delivered is too cold, the infant may be forced to use crucial energy it needs to survive to try and maintain its body temperature, resulting in stress on the body and a high chance of complications for the infant.

Additionally the required flow rate for the mNICU is also crucial for the babies survival. If the flow rate isn't set high enough there becomes a risk of an unsterile environment for the infant. Risk of dangerous carbon dioxide levels becomes high, as well as the possibility of airborne illnesses and contaminants entering the baby's environment. Greatly increasing the chances of infection and illness for the baby.

Finally, inconsistent or unsteady temperatures can also be fatal for infants. If provided temperatures can not remain steady the baby's body will become exhausted and suffer from metabolic stress trying to constantly micro adjust its internal body temperature. Additionally, inconsistent temperature control also puts a strain on the mNICU AHU system and can lead to premature breakdown and excessive wear on the NICUs components.